# Arrival

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Look at the letter tiles A and a. Point to one detail you notice.

| Letter tiles |
| --- |
| A a A a |

Point, sign, speak, draw, type or write. You may pass or pause. An adult can read or scribe your exact words. Plan in any language.

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Version _____



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## Starter

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Recall three ideas. Show one answer at a time.

1. What is a gene?
2. What do stem cells do in growth?
3. Where are chromosomes found in an animal cell?

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# Word page

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

| Word | Meaning | Example |
|---|---|---|
| allele | a version of a gene | A and a are alleles in our invented model. |
| genotype | the pair of alleles shown by letters | Aa is one genotype. |
| dominant | an allele shown by one copy in this simple model | A is dominant in the invented flower model. |
| recessive | an allele shown when two copies are present in this model | aa shows the recessive form. |
| Punnett square | a grid of possible allele pairs | Aa × aa has four cells. |

Use words or notes in your own language to help you learn these terms.



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# I do · full model

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Place A and a across the top. Put a and a at the side. Each box gets one allele from each parent.

| **Aa × aa · side / top** | **A** | **a** |
|---|---|---|
| a | Aa | aa |
| a | Aa | aa |

Two of four boxes are aa. This is a 1 in 2 chance for each offspring. It is not a fixed family count.

Invented flower model. The grid shows possible pairs. Show me: which allele can the aa parent pass on?

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# We do · missing step

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Complete the second row. Say which top and side alleles you used.

| **Aa × aa · side / top** | **A** | **a** |
|---|---|---|
| a | Aa | aa |
| a | ____ | ____ |

Chance of aa: ____ out of four.

One-minute reset: use your agreed pause, then return to one box.

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

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# We do · digital grid

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Complete the 2 × 2 grid for Aa × aa. Choose one allele from the top and one from the side for each box.

| Aa × aa · side / top | A | a |
|---|---|---|
| a | _____ | _____ |
| a | _____ | _____ |

You may point or direct an adult to enter your own choices. Check, explain and try again.

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## We do · choose one route

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Supported: Move one top tile and one side tile into one box. Point to the next pair.

Independent: Explain why the two aa boxes are half the outcomes.

Aspire: Predict the aa chance for Aa × Aa before you check a second grid.



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# We do 2 · hinge

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

New cross: Aa × Aa. What is the chance of aa for each offspring?

A. 1 in 4 for each offspring.

B. 1 in 2 for each offspring.

C. 3 in 4 for each offspring.

D. Certain for every single offspring.

Show your own choice. Give one reason. You may pass on speaking.



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# Independent · Supported

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Calculate the aa probability for Aa × aa. Evaluate an exact-count claim about real offspring.

Complete Aa × aa with allele tiles.

| **Aa × aa · side / top** | **A** | **a** |
|---|---|---|
| a | ____ | ____ |
| a | ____ | ____ |

Top ____ and side ____ make ____.

Choose one route for the full 8 minutes. Keep your own response. The one-minute reset is inside this time.

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# Independent · Independent

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Calculate the aa probability for Aa × aa. Evaluate an exact-count claim about real offspring.

Calculate and justify the aa chance from the grid.

| Aa × aa · side / top | A | a |
|---|---|---|
| a | ____ | ____ |
| a | ____ | ____ |

I count ____ aa boxes out of four, so ____.

Choose one route for the full 8 minutes. Keep your own response. The one-minute reset is inside this time.



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# Independent · Aspire

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Calculate the aa probability for Aa × aa. Evaluate an exact-count claim about real offspring.

Evaluate an exact-count claim about four real offspring.

| **Aa × aa · side / top** | **A** | **a** |
|---|---|---|
| a | ____ | ____ |
| a | ____ | ____ |

The grid shows a chance, not a promise, because ____.

Choose one route for the full 8 minutes. Keep your own response. The one-minute reset is inside this time.



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# Go deeper · faulty headings

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Compare aa chances for Aa × Aa and Aa × aa with a second grid.

Faulty headings audit: an Aa parent has been given A and A at the top. Correct both headings and explain the wrong probability.

| **Faulty Aa × aa · side / top** | **A** | **A** |
|---|---|---|
| a | Aa | Aa |
| a | Aa | Aa |

This is a deliberate error to repair. Use this task within the 8 minutes.



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# Optional tile model

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Kit: large A and a tiles, prepared grid and a tray.

Place one tile from each parent into each box. Say the pair. Collect all tiles after.

Risk: loose tiles could be mouthed or fall. Safe: use large tiles on a tray. Keep them out of mouths.

You can point at the printed grid instead. This is model rehearsal, not a core practical.



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## Review and improve

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Tell your partner what changed after feedback. Mark a real edit // and ask for a read-back ? if needed.

Keep your first answer. Hear the feedback. Choose one real change, or explain why to keep it. Ask for a read-back if it helps.

One-minute reset: use the agreed move, then return to one step.

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# Exit ticket · Supported

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

### 1. Show your model and explain one reason

Fill Aa × aa. Explain why two of four model boxes are aa, with no promise about real offspring.

### 2. Feedback and next step

My feedback was: _____

Next I need to check the top and side alleles before giving a chance.

My learning: I am learning to calculate genetic probability with a Punnett square and evaluate its limits.

Your teacher hears your answer and uses it to choose the next model, prompt or example.



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# Exit ticket · Independent

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

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# Exit ticket · Aspire

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

### 1. Show your model and explain one reason

Fill Aa × aa. Explain why two of four model boxes are aa, with no promise about real offspring.

### 2. Feedback and next step

My feedback was: _____

Next I need to check the top and side alleles before giving a chance.

My learning: I am learning to calculate genetic probability with a Punnett square and evaluate its limits.

Your teacher hears your answer and uses it to choose the next model, prompt or example.

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# My own working

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

Name: _____

Cross: Aa × aa

| **Aa × aa · side / top** | **A** | **a** |
|---|---|---|
| a | ____ | ____ |
| a | ____ | ____ |

My aa chance and reason: _____

My evidence that chance is not a fixed count: _____

Feedback and my next step: _____

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# Knowledge organiser

I can pair one allele from each parent.

I can find the aa chance in four squares.

I can evaluate why probability does not guarantee an exact count.

I am learning to calculate genetic probability with a Punnett square and evaluate its limits.

| Aa × aa · side / top | A | a |
|---|---|---|
| a | Aa | aa |
| a | Aa | aa |

One top allele plus one side allele makes one pair. Count two aa boxes out of four. The aa chance is one half for each offspring.

This is a simple model for one gene. A dominant allele is not stronger, better or more common. Each fertilisation is a new chance. Do not use real family traits or ask for personal or medical details.

| Word | Meaning |
|---|---|
| allele | a version of a gene |
| genotype | the pair of alleles shown by letters |
| dominant | an allele shown by one copy in this simple model |
| recessive | an allele shown when two copies are present in this model |
| Punnett square | a grid of possible allele pairs |

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